

SCENARIO 2

A Silesian Family Home

SCENARIO AT A GLANCE

Marek and Ania live with their two children in a detached house in Silesia. Marek works shifts in industry, Ania works in a hybrid model, and the children follow their own school and leisure schedules.

The household is busy and its daily routine changes depending on work shifts, school activities and whether Ania is working from home. Several appliances are often used at the same time, especially in the morning and after the family returns home.

The family wants to maintain the same level of comfort while better understanding when and why energy demand is highest and how it could be managed more efficiently.

What you need to do

- build a realistic historical hourly energy consumption profile for a minimum of 30 consecutive days,
- forecast energy demand for the next 24 hours, with the possibility of extending the forecast to 3 and 7 days,
- use real historical and forecast weather data,
- identify peak demand periods and explain changes in consumption,
- present the results in an application,
- evaluate how renewable energy and better timing of appliance use could reduce grid electricity consumption and household costs.

What you need

You will work with:

- the household profile and description of the residents' behaviour,
- indicative energy consumption parameters for household appliances and systems,
- an example household location for obtaining real historical and forecast weather data,
- your own clearly documented assumptions where additional information is required.

Context

Marek and Ania live with their two children, Kuba and Zosia, in a detached house in Silesia.

Marek works shifts in industry, while Ania combines office work with working from home. Kuba and Zosia attend school and spend part of their free time using computers, television and gaming devices.

The house combines an older building infrastructure with newer household appliances. The family's daily routine is busy and not always predictable. In the morning and evening, several appliances may operate simultaneously because household activities need to be completed within a relatively short period of time.

The family does not want to radically change its lifestyle. Instead, they want a solution that will help them forecast future energy demand, identify periods of particularly high consumption and find practical ways to use energy more efficiently without reducing everyday comfort.

Household Profile

4 people	2 adults and 2 school-age children
detached house in Silesia	one adult working shifts
one adult working in a hybrid model	older house combined with newer household appliances
electric heating or a heat pump	variable occupancy during the day
relatively high morning and evening activity	irregular weekly routine

Appliances and Indicative Energy Consumption Parameters

APPLIANCE / SYSTEM	INDICATIVE ENERGY CONSUMPTION
Heat pump / electric heating	1.5–3.5 kW during operation
Refrigerator	0.8–1.2 kWh/day
Kettle	2.0 kW, typically 3–5 min/use
Coffee machine	1.0–1.5 kW, typically 5–10 min/use
Oven	2.0–2.5 kW during operation
Electric / induction hob	1.5–3.5 kW depending on use
Washing machine	0.6–1.0 kWh/cycle
Dishwasher	0.8–1.2 kWh/cycle
Television	0.08–0.15 kW during operation
Gaming console	0.10–0.20 kW during operation
Desktop computer	0.10–0.30 kW during operation
Laptop	0.04–0.08 kW during operation
Wi-Fi router	0.008–0.015 kW, continuous operation
Lighting	0.08–0.20 kW total when active
Phone / tablet charging	0.005–0.02 kWh/device per charge
Standby devices	0.03–0.07 kW combined



EXAMPLE HOUSEHOLD LOCATION

Katowice, Poland

The values provided above are indicative and are intended to serve as input parameters for the challenge. Participants may make reasonable additional assumptions where necessary, provided that these assumptions are clearly documented in the final solution.

Participants should use real historical and forecast weather data for the indicated location.

Challenge

Your task is to create a solution that forecasts the household's energy demand and optimises that demand in terms of cost.

1 Historical Energy Consumption Profile

Based on the provided scenario, prepare a realistic historical electricity consumption profile for the household.

The historical profile should cover **a minimum of 30 consecutive days with hourly resolution.**

The profile should:

- reflect the residents' daily routines,
- include appliance and system operation,
- account for the impact of outdoor temperature,
- include realistic differences between individual days,
- reflect periods when different household members are at home,
- reflect the behavioural variability described in the scenario.

The historical profile will serve as input data for the next part of the task.

2 Energy Consumption Forecast

Prepare a model that forecasts the household's hourly energy demand.

The solution must generate a forecast for:

- the next 24 hours.

The solution should also support forecasts for:

- the next 3 days,
- the next 7 days,

depending on data availability.

The forecast should account for predicted weather conditions and expected household occupancy.

3 Application

Prepare an application that presents the system's results.

The application must display:

- forecast hourly energy consumption,
- forecast total energy consumption for the selected period,
- outdoor temperature forecast,
- hours with the highest expected demand,
- a brief explanation of the expected increase or decrease in consumption.

The user should be able to view a forecast for at least the next 24 hours.

05

Renewable Energy Simulator

The solution should include a module for assessing the impact of an investment in renewable energy sources on the household.

The module should allow the user to select the installation capacity in kWp.

For the purpose of the challenge, it may be assumed that the house has suitable conditions for a rooftop photovoltaic installation.

For the selected option, the application should present:

01

estimated annual energy production,

02

**the share of self-generated energy
in covering household demand,**

03

**estimated reduction in electricity
purchased from the grid,**

04

estimated savings,

05

indicative payback period for:

A

a renewable energy installation
without changing current habits,

B

a renewable energy installation
combined with shifting selected
energy-consuming activities
to more favourable periods.

The module should enable comparison of at least several installation capacity options.

We provide common assumptions for electricity tariffs and PV costs for all teams in the shared attachment. For any other assumptions, such as renewable energy production, you are free to choose your own values. Just state them clearly in your solution.

Description of Residents' Behaviour

5:30–8:00

The family's morning routine depends partly on Marek's work shift.

Marek may be preparing to leave for work, already be at work, or return home after a night shift.

At the same time:

- lighting is used,
- the kettle and coffee machine operate,
- breakfast is prepared,
- the children prepare for school,
- electronic devices may be charged,
- heating maintains a comfortable indoor temperature,
- the washing machine may occasionally be started.

Several appliances may operate simultaneously.

8:00–14:00

The children are normally at school.

Ania may:

- work from the office,
- work from home.

Marek's presence depends on his shift schedule.

When somebody is at home, electricity consumption may include:

- laptop or computer use,
- television or radio,
- food preparation,
- washing machine use,
- heating,
- lighting.

On some days, the house may therefore remain active throughout the morning and early afternoon.

14:00–17:30

The children return home from school and household activity gradually increases.

They may:

- watch television,
- use computers,
- play games,
- charge electronic devices,
- prepare snacks.

Depending on the day, Ania may still be working from home or return from the office, while Marek's presence continues to depend on his work shift.

17:30–21:30

This is often the most energy-intensive period.

During this time:

- dinner is prepared,
- the oven or hob may be used,
- the dishwasher may operate,
- the washing machine may be used,
- television and gaming devices are active,
- computers and electronic devices are used,
- lighting use increases,
- heating maintains comfortable indoor conditions.

Several appliances may operate at the same time, creating a significant evening peak. This pattern reflects the busy family routine described in the original persona.

21:30–5:30

Most household activity decreases.

The following remain active:

- refrigerator,
- heating system,
- Wi-Fi router,
- standby devices,
- phone and tablet charging.

However, Marek's shift work means that electricity use may occasionally occur at unusual hours.

Behavioural Variability

The household profile is not identical every day.

The following should be taken into account:

- Marek works different shifts, which affects both departure and return times,
- Ania works from home on selected days,
- the children's schedules differ between school days and weekends,
- meal times may shift,
- washing and dishwasher use vary between days,
- household members may leave the house for several hours,
- heating-related energy consumption changes with weather conditions,
- weekends have a different occupancy profile,
- unusual activity may occur early in the morning or late at night because of shift work,
- several appliances may operate simultaneously, creating short periods of high demand.

Required Outcome

At the end of the challenge, the solution should include:

- | | |
|---|--|
| ☐ 1 a historical hourly energy consumption profile covering a minimum of 30 consecutive days, | ☐ 2 an energy consumption forecasting model, |
| ☐ 3 an hourly forecast for the next 24 hours, | ☐ 4 the ability to present forecasts for the next 3 and 7 days, |
| ☐ 5 an application presenting the results, | ☐ 6 identification of the hours with the highest expected demand, |
| ☐ 7 a brief explanation of changes in forecast energy consumption, | ☐ 8 a simulator showing the impact of a renewable energy investment on the household's energy balance and costs. |

Final Objective

The solution should make it possible to answer the following questions:

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- Q1** What energy demand can be expected in this household over the next 24 hours, 3 days, and 7 days?
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- Q2** During which hours will energy demand be highest?
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- Q3** What changes in household habits could be proposed to optimise the use of renewable energy and electricity from the grid?
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- Q4** How could the selected renewable energy investment affect the amount of electricity purchased from the grid, household costs, and the investment payback period?
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